

Retrieval-Augmented Few-Shot Prompting vs. LoRA Fine-Tuning for Issue Report Classification: An Empirical Study

Anonymous author

Anonymous affiliation

Abstract

Background. Issue report classification (IRC) accelerates modern software maintenance. State-of-the-art automated IRC methods fine-tune Large Language Models (LLMs) with Low-Rank Adaptation (LoRA), which is computationally expensive and must be repeated to incorporate newly labeled issue reports or whenever the model is swapped.

Aims. We empirically investigate whether retrieval-augmented few-shot prompting offers a practical alternative to LoRA fine-tuning for IRC.

Method. Given a query issue, we retrieve its top- k nearest neighbors from an index of classified issues, and propose three methods: (i) VOTAG, a similarity-weighted vote on the neighbors' labels, (ii) RAGTAG, which presents the neighbors as classified examples in a few-shot prompt to an LLM, and (iii) BRAGTAG, a RAGTAG variant with a more balanced retrieval scheme. We evaluate all three against a LoRA fine-tuning baseline on an 11-project benchmark, in both *project-specific* and *project-agnostic* data scope settings. Experiments use Qwen2.5-Instruct at four sizes.

Results. With project-specific data, RAGTAG outperforms fine-tuning by 0.021–0.040 macro F_1 across all model sizes. Fine-tuning on cross-project data beats project-specific RAGTAG by only 0.029 aggregate macro F_1 despite $11\times$ more labeled data. The balanced retrieval approach, BRAGTAG closes this performance gap, achieving statistical equivalence with fine-tuning when using a retrieval fallback for invalid LLM outputs. Averaged across model sizes, Retrieval-augmented few-shot prompting requires 33% less peak GPU memory than fine-tuning.

Conclusions. Retrieval-augmented few-shot prompting is a practical alternative to LoRA fine-tuning for IRC, particularly when hardware resources are constrained, labeled data is limited, or the deployment has frequent model or data updates.

2012 ACM Subject Classification Software and its engineering → Software notations and tools; Computing methodologies → Natural language processing

Keywords and phrases Issue classification, Retrieval-augmented generation, Fine-tuning, Large language models, Mining software repositories

Digital Object Identifier 10.4230/LIPIcs...

1 Introduction

Issue Tracking Systems provide a structured way to communicate concerns related to the software project by allowing users to submit issue reports [14]. Issue reports can be classified with labels to help with triage. Effective issue report classification (IRC) results in faster issue resolution [35], and is crucial for many software maintenance tasks, such as issue prioritization [10], issue assignment [50, 30], and developer onboarding [42, 44, 5]. However, studies show that labels are rarely applied in issue creation [3, 24, 9, 31]. As a result, project maintainers must manually classify the issues for triage, which is laborious and time-consuming [19].

To address this problem, numerous automated methods have been proposed around supervised learning over three established issue report categories: bug, feature, and question [3,



© Anonymous author(s);

licensed under Creative Commons License CC-BY 4.0

Leibniz International Proceedings in Informatics

LIPICS Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany

31, 28, 5, 4]. Early work relied on classic machine learning (ML) techniques [3, 31, 19, 24], while more recent studies fine-tune transformer models on labeled issue reports [28, 29, 45, 7, 11, 48, 22]. These approaches require a substantial amount of training data and often struggle with minority classes [28, 29, 11, 23, 46]. The current state-of-the-art methods fine-tune Large Language Models (LLMs) using Low-Rank Adaptation (LoRA [25]) [23, 5, 4]. However, fine-tuning LLMs is computationally expensive and must be repeated whenever the model is swapped or to incorporate new labeled issues that accumulate in the project [19, 31].

Prior work shows that placing a few demonstration examples in the prompt can boost an LLM’s performance on specialized tasks [6, 34, 39, 27, 8] through in-context learning [8]. Since few-shot prompting enhances performance without model training or weight updates, it has the potential to serve as a cost-efficient alternative to fine-tuning LLMs for IRC. Few-shot performance, however, depends on the choice of in-context examples. Studies show that using the query’s retrieved nearest neighbors as in-context examples yields strong few-shot performance [36, 52]. Additionally, the nearest neighbors tend to share the same label as the query [15], making them suitable in-context examples. The combination of these insights leads to retrieval-augmented few-shot prompting, where the classified issue reports most similar to a query are retrieved and used as its in-context examples. Yet both a detailed empirical evaluation of this approach for IRC and its comparison against fine-tuning remain underexplored. We aim to address both gaps in this study.

An important factor in retrieval-augmented few-shot performance is the number of provided examples: too few may not provide enough context and too many can add noise [36]. Since the in-context examples are the query’s nearest neighbors, they can classify the query under the standard k -nearest-neighbor (KNN) framework [15, 18]. Therefore, the point where adding neighbors stops improving the retrieval-only classification informs an upper bound for the number of in-context examples. Additionally, retrieval-only IRC provides a non-parametric baseline that isolates the LLM’s contribution from that of retrieval in retrieval-augmented few-shot prompting. To our knowledge, no prior study has explored such a baseline for IRC. Therefore, we pose our first research question:

RQ1. How effectively can retrieval alone classify issue reports, and at what point does adding more retrieved neighbors stop helping? We answer this question by applying a standard distance-weighted KNN classification [15, 18]. Given a query issue, its top- k most similar issue reports (nearest neighbors) are first retrieved from an embedded index of existing issue reports. Next, the label is predicted with a similarity-weighted vote over the neighbors. We evaluate this method over a wide range of top- k values to measure its peak performance and identify the point where additional neighbors are not beneficial (best k value). With the best k value established, we ask:

RQ2. How well does retrieval-augmented few-shot prompting classify issue reports, and how does performance vary with the number of in-context examples? We answer this question by presenting the same top- k nearest issue reports as classified examples in a few-shot prompt, instructing the model to predict a label for the query issue. $k=0$ results in a zero-shot LLM baseline similar to prior work [12]. We evaluate this approach with different model scales over a range of k values bounded by RQ1.

RQ2 evaluations reveal that retrieval-augmented few-shot has the weakest performance on the question class, and question issues are frequently misclassified as bugs. Zero-shot prompting results indicate that the models are already biased towards question-to-bug misclassifications even before the examples are presented. We hypothesize that bug-labeled neighbors retrieved for true-question queries reinforce this misclassification. This motivates our next research question:

RQ3. Can a balanced retrieval intervention reduce question-to-bug misclassification and improve retrieval-augmented few-shot performance? We answer this question by exploring a more balanced retrieval technique that excludes bug-labeled neighbors when they outnumber question-labeled ones by a margin in the top- k . We evaluate it over the same k values and model scales used for RQ2.

For brevity, we refer to the retrieval-only classifier (RQ1) as **V**oting-based **T**AGging (VOTAG), the retrieval-augmented few-shot approach (RQ2) as **R**etrieval-**A**ugmented **G**enerative **T**AGging (RAGTAG), and its balanced variant (RQ3) as **B**alanced RAGTAG (BRAGTAG). We compare these methods against the established LoRA [25] fine-tuning baseline [23, 5, 4] to answer our final research question:

RQ4. How do retrieval-based methods compare against LoRA fine-tuning for IRC in classification performance and computational cost? We conduct this comparison across the Qwen2.5-Instruct model family at 3B, 7B, 14B, and 32B parameters [51] to analyze the effect of model scale without architecture variance. We use Heo et al.’s [23] 11-project dataset for our experiments and adopt its two evaluation settings: a *project-specific* (PS) configuration where retrieval/training is over each project’s individual data split in isolation, and a *project-agnostic* (PA) configuration where retrieval/training is over data from the 11 projects pooled together.

Results show that retrieved few-shot examples consistently improve IRC performance. Even a single retrieved example outperforms the zero-shot baseline across all model sizes (Section 5.2). LLM reasoning also consistently improves over pure retrieval: every RAGTAG configuration outperforms VOTAG’s best result. Notably, however, VOTAG comes within 0.018 macro F_1 of Qwen-3B’s zero-shot (Section 5.1).

Additionally, retrieval-augmented few-shot prompting is competitive with fine-tuning. Using only each project’s own labeled issues (PS), RAGTAG outperforms fine-tune by 0.021–0.040 macro F_1 on all model sizes (Section 5.4). Fine-tune only catches up when trained on $11\times$ more labeled data via cross-project pooling (PA), and even then RAGTAG remains competitive: aggregated across all model sizes, fine-tune PA leads RAGTAG PS by 0.029 macro F_1 , and the two methods are statistically tied at Qwen-3B and Qwen-32B. BRAGTAG closes this gap by reducing the question-to-bug misclassification rate without sacrificing performance on other labels (Section 5.3). The aggregate (BRAGTAG – fine-tune) macro F_1 difference is -0.010 . Applying a VOTAG fallback to invalid LLM outputs across all methods tightens this to statistical equivalence, passing two one-sided tests (TOST) at $\delta=0.01$.

Retrieval-augmented few-shot offers deployment advantages as well. Averaged across the four model sizes, RAGTAG and BRAGTAG require 33% less peak GPU memory than fine-tuning (Section 5.4). Additionally, incorporating newly labeled issues and swapping the underlying model mid-production do not require retraining cycles.

Overall, this study demonstrates retrieval-augmented few-shot’s practicality in IRC, particularly when hardware resources are constrained, labeled data is limited, or the deployment requires frequent model or data updates.

The main contributions of this paper are the following:

- Establishing a retrieval-only, non-parametric baseline for IRC that also serves as a fallback mechanism for invalid LLM predictions (VOTAG).
- Conducting the first systematic evaluation of retrieval-augmented few-shot prompting for IRC across different LLM scales and numbers of in-context examples (RAGTAG).
- Proposing a balanced retrieval technique for few-shot prompting that reduces question-to-bug misclassifications without degrading performance on the other labels (BRAGTAG).

Empirically comparing these proposed methods against the established LoRA fine-tuning baseline on both classification performance and computational cost, in two data scope settings (PS and PA). We find that retrieval-augmented few-shot prompting is competitive with fine-tuning while using $11\times$ less labeled data per project and 33% less peak GPU memory.

The remainder of this paper is organized as follows: Section 2 reviews prior work. Section 3 explains our proposed methods and the fine-tuning baseline. Section 4 details the experimental setup. Section 5 explains our evaluations and reports the results. Section 6 discusses failure modes, practical implications, and future work. Section 7 addresses the potential threats to validity. Finally, Section 8 concludes the paper.

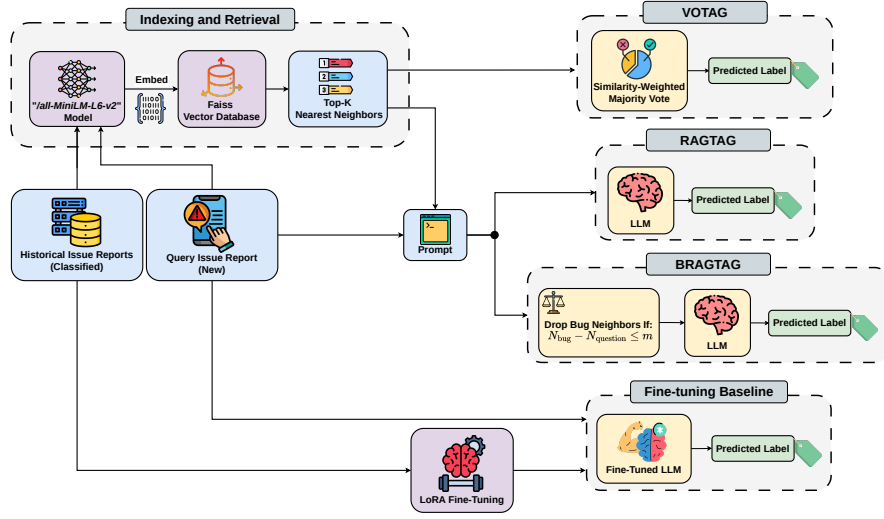
2 Related Work

Automated issue report classification (IRC) Early IRC studies employed data mining and ML classifiers [3, 31, 19]. For example, Antoniol et al. [3] leveraged a naive Bayes classifier to distinguish bugs from other issue reports. However, these methods yielded limited performance due to their shallow semantic understanding of issue report text [23]. To overcome this limitation, more recent work has adopted BERT-like transformer models for their stronger contextual understanding [11, 7, 45, 28, 29]. These studies typically fine-tune the models on historical issues in a supervised setting. For instance, Izadi et al. [28] fine-tuned a RoBERTa [38] model to classify issues into *Bugs*, *Features*, or *Questions*. Similarly, Trautsch et al. [45] trained a seBERT [47] model to classify issues into the same three categories. Although transformer-based approaches achieve strong results, they rely on large amounts of training data, which limits their generalizability to settings where labeled data is scarce [23, 28, 29, 11]. These challenges have motivated the exploration of LLMs for IRC. Their extensive pretraining allows these models to be applied with little or no task-specific training, reducing the dependence on the large labeled datasets that supervised classifiers require. [5, 4, 14, 12, 13]. However, state-of-the-art LLM-based methods do not leverage this property. Instead, they fine-tune the LLM with Low-Rank Adaptation (LoRA [25]) on labeled issue datasets to achieve their best reported performance [23, 5, 4]. Although fine-tuning is effective, it reintroduces the dependency on labeled training data, remains computationally expensive, and must be repeated whenever the underlying model is swapped or as newly labeled issues accumulate in the project [19, 31].

Retrieval-augmented few-shot prompting Studies show that few-shot prompting can improve LLM performance in specialized tasks due to in-context learning [6, 34, 39, 27, 8]. Few-shot prompting has also been applied in IRC. Colavito et al. [12, 14] randomly selected one or two labeled examples from their dataset and presented them in a few-shot prompt. Few-shot performance, however, is sensitive to the quality of in-context examples. Research shows that using retrieval techniques to select the most semantically relevant in-context examples can outperform random selection [36, 52]. Since retrieval-augmented few-shot prompting can enhance performance without the overhead of training, it can be a promising alternative to LoRA fine-tuning IRC baseline. To the best of our knowledge, however, this method has not been systematically and empirically evaluated for IRC.

3 Approach

In this section, we describe the intuition behind, and the full implementation details of our three proposed retrieval-based IRC methods (VOTAG, RAGTAG, and BRAGTAG) as



■ **Figure 1** Overview of the four IRC approaches evaluated in this paper: VOTAG, RAGTAG, BRAGTAG, and the LoRA fine-tuning baseline.

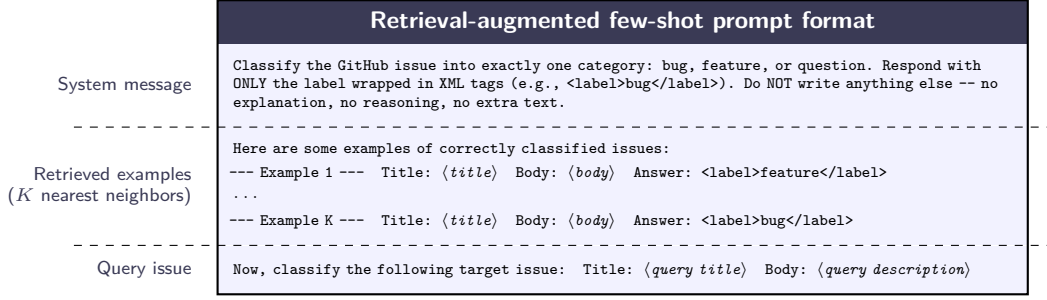
well as the faithful reproduction steps of the established LoRA fine-tuning baseline [23, 5]. Figure 1 illustrates all four approaches to support the explanations. The remainder of the section is organized as follows: Section 3.1 explains the overall IRC objective. Section 3.2 details the indexing and retrieval pipeline shared by our three proposed retrieval-based methods. Section 3.3 presents VOTAG, our non-parametric method of classifying the query directly from its nearest neighbors. Section 3.4 describes RAGTAG, our few-shot prompting technique that presents the nearest neighbors as classified examples in the prompt. Section 3.5 elaborates on BRAGTAG’s balanced retrieval intervention on RAGTAG. Finally, Section 3.6 details the faithful reproduction of the fine-tuning baseline.

3.1 Problem Formulation

Given a query issue report, we classify it into only one of the three labels: *bug*, *feature*, or *question*. Only the issue’s title and description text are used for classification with no other signals from the issue tracker. The defined label space is adopted directly from prior IRC studies [4, 5, 28, 31, 3].

3.2 Indexing and Retrieval

To prepare the issues for retrieval based on textual similarity, we first embed their textual content (i.e., *title* + *description*). We do not embed the issue labels. Instead, we retain them as metadata associated with each embedded issue. To generate the embeddings, we use the `all-MiniLM-L6-v2` model from the Sentence-Transformers library [41]. We selected this model due to its lightweight architecture, fast performance, and effectiveness on issue report text [26]. The embeddings are then indexed using the Faiss vector database library [17], which provides efficient high-dimensional similarity retrieval. At the retrieval stage, the query issue is first embedded with the same model. Next, Faiss calculates the cosine similarity of indexed issue reports to the query, and returns the top-K nearest neighbors, ordered by descending similarity.



■ **Figure 2** The retrieval-augmented few-shot prompt structure used for RAGTAG and BRAGTAG.

3.3 VOTAG

VOTAG is our proposed non-parametric retrieval IRC method. VOTAG uses the retrieval pipeline described in Section 3.2 to retrieve the top- K nearest neighbors of the query issue report, then predicts the final label with a similarity-weighted vote on their labels. This approach follows the standard K -nearest-neighbor classification setup [15], with neighbors weighted by their similarity to the query [18].¹ Formally, for a query issue q with retrieved neighbors $\{(n_i, l_i, s_i)\}_{i=1}^K$, where n_i is the i -th nearest neighbor with label l_i and cosine similarity s_i to q , the predicted label $\hat{y}$ is given by:

$$\hat{y} = \arg \max_{c \in \mathcal{L}} \sum_{i=1}^K s_i \cdot \mathbf{1}[l_i = c], \quad (1)$$

where $\mathcal{L}$ is the label set, and $\mathbf{1}[l_i = c]$ is the indicator function that equals 1 when neighbor i has label c and 0 otherwise. For each label VOTAG sums the cosine similarities of the neighbors that carry that label, and the $\arg \max$ operator returns the label with the largest total sum. VOTAG breaks ties by returning the label of the most similar neighbor in the tied classes.

3.4 RAGTAG

RAGTAG is our proposed retrieval-augmented few-shot prompting approach for IRC. RAGTAG uses the retrieval pipeline described in Section 3.2 to retrieve the top- K nearest neighbors to the query issue, and present them as classified examples to the LLM in a few-shot prompt. This enables a more accurate classification, based on in-context learning [8]. The choice to present the top- K nearest neighbors as examples is based on two insights from prior research: First, retrieval-based example selection outperforms random selection [36]. Second, the most similar neighbors tend to share the same labels [15]; therefore, they are informative for the classification.

We create the few-shot prompt in three parts as shown in Figure 2. First, the system message frames the classification task and the label space. Second, the top- K nearest neighbors are listed in retrieval order (descending similarity), each formatted as *title + description + label*. Finally, the query issue is appended to the end of the prompt, ensuring it receives high attention from the LLM [37]. $K = 0$ produces a zero-shot prompting baseline, similar to prior work [12].

¹ Squared-similarity weighting and uniform majority were also evaluated, but were outperformed by similarity-weighted voting.

We evaluate RAGTAG at top-K values in $\{0, 1, 3, 6, 9, 12, 15\}$. Details for the K value grid selection are provided in Sections 5.1 and 5.2. The maximum prompt sequence length is set to 8,192 tokens.² Roughly 30% of the context quota is reserved for the query issue and 70% for the neighbors. If the neighbors exceed their overall quota, they are truncated proportionally based on their length (longer neighbors get more truncation). The query issue gets truncated if it exceeds its budget as well. Truncation happens on the right side of the description text, preserving all titles and labels (if any). The LLM is instructed to generate the predicted label wrapped in XML tags (e.g. `<label>bug</label>`). The labels of the presented examples also follow the same format. This technique enables easy detection of the predicted label from the LLM’s generation, and reduces variance in LLM outputs caused by prompt format sensitivity [43]. We use a regular expression to extract the predicted label. Unparseable outputs are marked as *invalid*. Finally, models are loaded with Unsloth³ For memory efficiency, and the Temperature is set at 0.1 for deterministic classifications. [AH: Is this a good rationale? When you set a number, reviewers often ask for the rationale.]

3.5 Balanced RAGTAG (BRAGTAG)

Our empirical evaluations in (Section 5.2) reveal that RAGTAG has the worst performance on the question label, and it frequently misclassifies true-question queries as bugs. Furthermore, LLM zero-shot results also show that the models are already biased towards predicting bug for true-question queries even before the examples are presented. Based on these insights, we hypothesize that presenting bug-labeled examples for true-question queries likely reinforces the question-to-bug misclassification trend. To address this, we propose a more balanced retrieval intervention, called balanced RAGTAG (BRAGTAG), that removes the bug-labeled neighbors whenever the query is a suspected question based on its nearest neighbors’ label distribution. (Section 5.2) shows that true-question queries retrieve almost balanced bug and question neighbors. Due to this balanced distribution, BRAGTAG removes the bug-labeled examples from the prompt whenever $N_{\text{bug}} - N_{\text{question}} \leq m$, where N_{label} is the count of neighbors with that label in the top- k and m is a small positive margin. The margin is empirically calibrated on a holdout validation set. The empirical motivations for BRAGTAG and its margin calibration are fully detailed in (Section 5.2). Other than the balanced retrieval intervention, BRAGTAG inherits RAGTAG’s full setup and configurations.

3.6 LoRA Fine-Tuning Baseline

Recent IRC approaches have achieved strong performance by supervised fine-tuning of LLMs [23, 5, 4], using Low-Rank Adaptation (LoRA) [25]. In these studies, the model is trained on labeled issue reports and predicts the labels for unseen test issues. Each training example is formatted as an instruction prompt which includes the issue’s title and description as input, and its label as output. We adopt the instruction-style prompt template used in these prior LoRA fine-tuning studies [23, 5] verbatim, rather than designing our own, so that the baseline faithfully reproduces their established methodology. At inference, the trained model receives the test issue in the same format, and predicts the empty label field. Regular expressions are used to extract the prediction from the output, and unparseable outputs are marked as *invalid*. We use this established methodology as our fine-tuning baseline.

² Maximum prompt sequence length is selected empirically. 8,192 tokens offered the best trade-off between classification accuracy and inference speed across our preliminary experiments.

³ <https://unsloth.ai>

278 Training hyperparameters follow prior work [23, 5]: The LoRA rank and alpha are set to
 279 16, with a learning rate of 2×10^{-4} and the paged AdamW 8-bit optimizer. Training runs
 280 for 3 epochs with a per-device batch size of 1 and gradient accumulation steps of 16. The
 281 maximum prompt sequence length is set to 2,048 tokens, since 94.9% of the test issues in the
 282 dataset will not be truncated at this length. Issues that exceed this length get truncated
 283 from the right side of their description text to fit. Similar to RAGTAG and BRAGTAG, the
 284 models are loaded with Unsloth and the Temperature is set at 0.1.

285 4 Experimental Setup

286 **Dataset and Settings.** We use the dataset introduced in Heo et al. [23] for our experiments.
 287 The dataset has a total of 6,600 issue reports gathered from eleven active open-source projects:
 288 `ansible`, `bitcoin`, `dart-lang`, `roslyn`, `react`, `flutter`, `kubernetes`, `TypeScript`, `vscode`,
 289 `opencv`, and `tensorflow`. Each project has 600 issue reports with an equal distribution
 290 across the labels (*bug*, *feature*, *question*). To support their LLM fine-tuning method, Heo et
 291 al. divided the dataset into 50/50 train and test splits, stratified by project and label. This
 292 results in 300 train and 300 test issue reports per project.

293 We also adopt Heo et al.’s project-specific (PS) and project-agnostic (PA) configurations
 294 to examine how data scope affects IRC performance. In the PS setting, each project’s train
 295 and test data are used in isolation, resulting in eleven different train-test pairs. In the PA
 296 setting, the train splits of all projects are merged into a single training set of 3,300 issue
 297 reports. In both configurations, RAGTAG and VOTAG only use the training splits as their
 298 retrieval index to ensure fair comparison with the fine-tuning baseline.

299 **Models.** We use the Qwen2.5-Instruct model family [51] in our evaluations with four
 300 different parameter sizes: 3B, 7B, 14B, and 32B. Qwen2.5 is an open-source LLM with
 301 strong performance on general-purpose benchmarks. Using the same model at different
 302 parameter sizes isolates the effect of model scale from model architecture across LLM-based
 303 IRC approaches.

304 **Evaluation Metrics.** We report the per-class F_1 and the macro-averaged F_1 over the three
 305 classes (*bug*, *feature*, *question*). The macro F_1 is calculated over the total 3,300-issue
 306 test set in all settings. We also report the rate of outputs that fail to parse to a valid label
 307 (*invalid rate*). Invalid outputs are counted as incorrect predictions in all F_1 computations.
 308 For statistical significance checks, we report paired bootstrap 95% confidence intervals on
 309 the macro F_1 differences between methods.

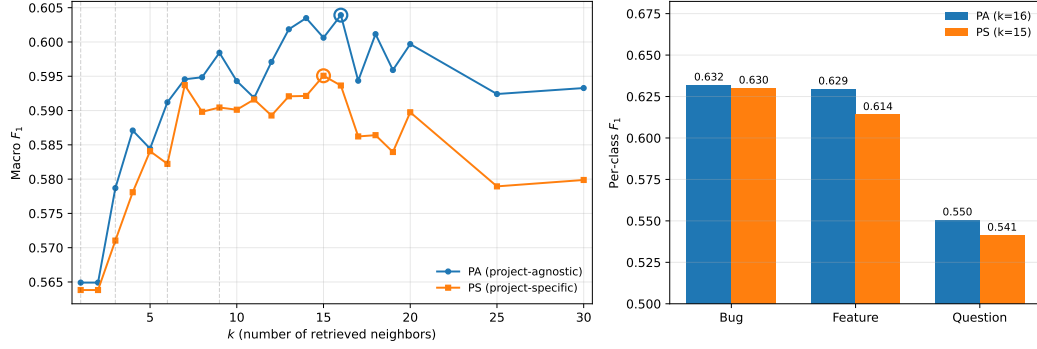
310 **Hardware Setup.** The experiments were conducted on a machine with one NVIDIA L40
 311 GPU (48 GB VRAM), 8 allocated CPU cores from an Intel Xeon Scalable processor, and
 312 64 GiB of system RAM.

313 5 Evaluations

314 In this section, we present the results of our experiments and answer the four research
 315 questions posed in Section 1, one per subsection.

316 5.1 RQ1: Effectiveness of Retrieval-Only IRC

317 We evaluate VOTAG’s performance on both PA and PS settings at k values 1 to 30. The
 318 resulting macro F1 scores are shown in Figure 3. VOTAG’s best macro F1 is 0.595 ($k=15$)



■ **Figure 3 (left)** VOTAG macro F_1 as a function of k in both PS and PA settings. Circles mark peak performance. **(right)** Per-class F_1 at each setting’s best k ($k=15$ for PS, $k=16$ for PA).

in PS and 0.604 ($k=16$) in PA. When adding neighbors beyond the peak, PS loses 0.015 and PA loses 0.011 by $k=30$.

PA consistently outperforms PS at every k . However, the performance gap is marginal: averaging 0.007 macro F_1 , with a maximum of 0.015 at $k=18$. This similar performance is due to the large overlap of the top- k neighbors in both settings. Specifically, PA and PS share 84–90% of the top- k neighbors across $k \in [3, 30]$. Therefore, even when using the full 3,300-issue retrieval index, most of the retrieved neighbors are from the same project.

Question consistently has the weakest performance among the three labels at every k in both settings. PS per-class F_1 at best k is 0.630 for bug, 0.614 for feature, and 0.541 for question. The corresponding PA values are 0.632, 0.629, and 0.550 (Figure 3 right panel). Additionally, questions are more frequently misclassified as bugs: 32% of question issues are predicted as bugs in both settings, while 18% (PS) and 17% (PA) are predicted as features. Since VOTAG predicts the label with the most accumulated similarity weight, this misclassification asymmetry indicates that question issues are more similar to bugs than to features in the embedding space.

Answer to RQ1: Retrieval alone establishes a non-parametric baseline with a peak macro F_1 of 0.604 and bounds the useful number of in-context examples at $k \approx 15$.

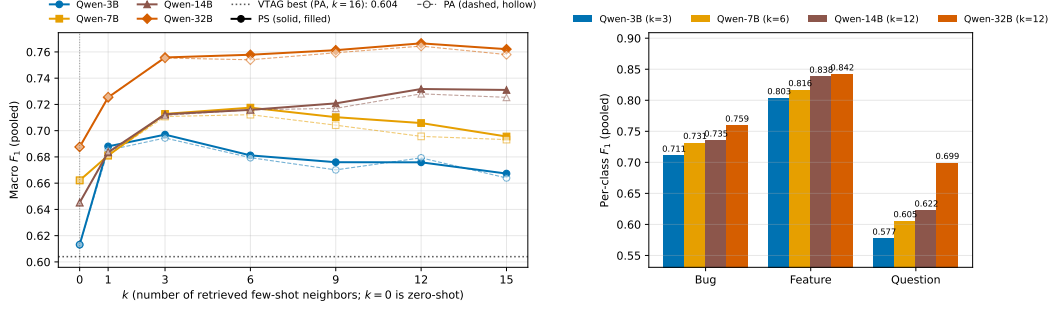
5.2 RQ2: Effectiveness of Retrieval-Augmented Few-Shot IRC

We evaluate RAGTAG at k values $\{0, 1, 3, 6, 9, 12, 15\}$ on both PA and PS settings, where $k=0$ is the zero-shot baseline [12]. k value selection is informed by VOTAG’s peak performance around $k=15$ in the previous section. Figure 4 (left panel) shows the macro F_1 of every model across this grid for both settings.

PS marginally outperforms PA at almost every (model, $k \geq 1$) pair evaluated (zero-shot is identical between settings since no retrieval is performed). The close performance gap is due to the same top- k neighbor overlap observed in the VOTAG analysis. The marginal improvement is significant because PS uses $11 \times$ less retrieval data per project: each PS index covers only its own 300 retrieval issues, while PA uses the full 3,300 across all 11 projects. Given this practical advantage, we conduct the rest of our RAGTAG analysis on the PS setting and report PS numbers throughout this section.

The best k grows with model size: Qwen-3B peaks at $k=3$ (macro $F_1 = 0.697$), Qwen-7B

XX:10 Retrieval-Augmented Few-Shot vs. Fine-Tuning for IRC



■ **Figure 4 (left)** RAGTAG macro F_1 as a function of k for all four Qwen sizes. $k=0$ is the zero-shot baseline. The dotted horizontal line marks VOTAG’s best (0.604, PA $k=16$). **(right)** Per-class F_1 for each model size at its best RAGTAG-PS configuration.

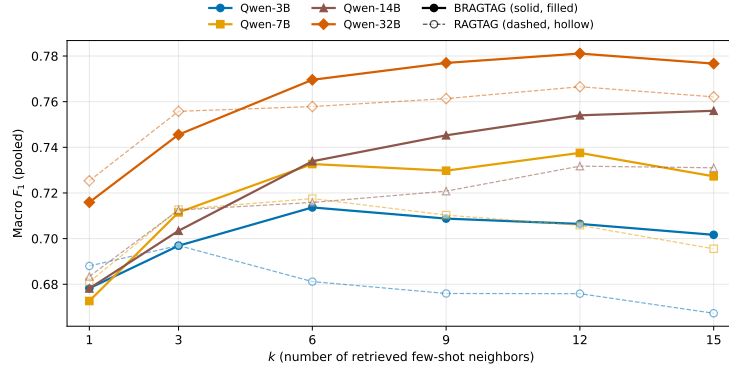
at $k=6$ (0.718), and both Qwen-14B and Qwen-32B at $k=12$ (0.732 and 0.767 respectively). This shows that larger models can use the context from additional examples more effectively. However, adding neighbors past $k=12$ has no benefit and slightly degrades performance: every model loses 0.001–0.015 macro F_1 going from $k=12$ to $k=15$.

Every RAGTAG configuration with $k \geq 1$ outperforms the zero-shot baseline. At each model’s peak, the gap over zero-shot averages +0.076 macro F_1 and ranges from +0.055 (Qwen-7B) to +0.087 (Qwen-14B). This shows that LLMs benefit substantially from the most siClassification performance similar labeled examples for IRC.

Every RAGTAG configuration outperforms VOTAG’s best result (0.604 at PA $k=16$), with the smallest margin from Qwen-3B zero-shot (+0.009 F_1) and the largest from Qwen-32B PS at $k=12$ (+0.163 F_1). These results show that adding the LLM reasoning improves IRC on top of pure retrieval.

Similar to VOTAG, question consistently has the weakest performance among the three labels across all RAGTAG settings. Figure 4 (right panel) shows per-class F_1 at each model’s best k . Even the smallest gaps between question and the other labels, observed at Qwen-32B with $k=12$, are noticeable: question’s is macro F_1 0.060 below bug’s and 0.143 below feature’s. Similar to the VOTAG observations, questions tend to be more frequently misclassified as bugs. At zero-shot, across all four model sizes, 46–59% of true-question issues are predicted as bugs while only 5–16% are predicted as features. RAGTAG narrows this misclassification asymmetry but does not eliminate it: at each model’s best k , 26–36% of question issues are still predicted as bugs and only 9–15% as features.

These results show that the LLMs are already prone to misclassifying question issues as bugs, even before the top- k examples are used. Therefore, presenting bug-labeled examples for true-question query issues likely helps these misclassifications. Based on this intuition, we propose a more balanced retrieval technique as an intervention that removes the bug-labeled neighbors whenever the query is a suspected question. We base the suspicion signal on the nearest neighbors’ label distribution and calibrate it without including the test set. From each project’s total 300 training issues, we assign 30 issues stratified by label as queries and 270 as the retrieval index, mirroring the PS setting. On this validation split, true-question queries retrieve almost balanced bug and question examples: averaged across $k \in [1, 15]$, the mean of $N_{\text{bug}} - N_{\text{question}}$ is -1.36 . We therefore treat a query as a suspected question when $N_{\text{bug}} - N_{\text{question}} \leq m$, where N_{label} is the count of neighbors with that label in the top- k



■ **Figure 5** Macro F_1 as a function of k across all model sizes for both RAGTAG and BRAGTAG (project-specific setting)

and m is a small positive margin. Sweeping $m \in \{1, 2, 3, 4, 5\}^4$ on the validation split and averaging across $k \in [1, 15]$, $m=3$ had a more favorable trade-off against the other values, firing for 89.1% of true-question and 56.7% of true-bug queries. We adopt $m=3$ and refer to the resulting method as **Balanced RAGTAG** (BRAGTAG).

Answer to RQ2: Every Retrieval-augmented few-shot prompting (RAGTAG) configuration with $k \geq 1$ consistently outperforms the zero-shot and VOTAG baselines. The best k grows with model size but plateaus within the $k \approx 15$ bound established by VOTAG. RAGTAG is weakest on the question class and frequently misclassifies questions as bugs, motivating a more balanced retrieval method.

5.3 RQ3: Effectiveness of the Balanced Retrieval method

Since the PS setting produced the best results for RAGTAG, we use PS as the default for evaluating BRAGTAG on the same $k \in \{1, 3, 6, 9, 12, 15\}$ grid. Figure 5 shows macro F_1 vs k for both methods across all four model sizes. BRAGTAG outperforms RAGTAG at every $k \geq 6$. At smaller $k \in \{1, 3\}$, BRAGTAG slightly underperforms by 0.001–0.010 macro F_1 , because the neighbor balancing intervention removes every retrieved example too often, practically reverting the query to zero-shot (40% of queries at $k=1$ and 18% at $k=3$).

BRAGTAG’s best k almost always increases compared to RAGTAG: Qwen-3B from $k=3$ to $k=6$, Qwen-7B from $k=6$ to $k=12$, Qwen-14B from $k=12$ to $k=15$, and Qwen-32B unchanged at $k=12$. When the trigger fires, BRAGTAG removes all bug examples from the top- k . Therefore, A higher initial k is needed to leave enough remaining context for the LLM to use. At each method’s best k , BRAGTAG outperforms RAGTAG on every model. The paired bootstrap 95% CI on the (BRAGTAG – RAGTAG) macro F_1 difference excludes zero in all four cells:⁵ Qwen-3B +0.017 [+0.003, +0.031], Qwen-7B +0.020 [+0.007, +0.032], Qwen-14B +0.024 [+0.014, +0.034], and Qwen-32B +0.014 [+0.007, +0.022]. Table 1 reports the full best configuration comparison.

The performance gain comes from an increase on the question and bug classes with minimal impact on feature. Table 1 reports per-class F_1 at each method’s best k . Question

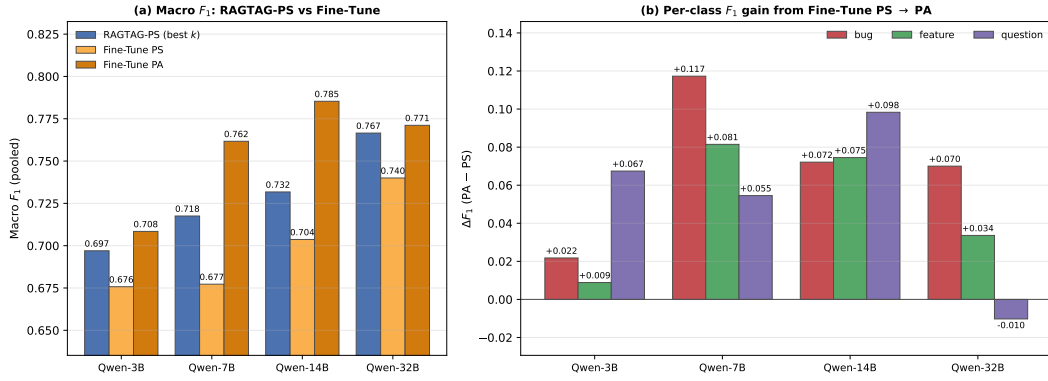
⁴ We cap the sweep at $m=5$, since it is the 95th percentile of $N_{\text{bug}} - N_{\text{question}}$ on true-question queries.

⁵ 1,000 paired bootstrap resamples per model.

XX:12 Retrieval-Augmented Few-Shot vs. Fine-Tuning for IRC

■ **Table 1** RAGTAG vs BRAGTAG at each model’s best k (PS, pooled).

Model	Method	k^*	Macro F_1	F_1^{bug}	F_1^{feat}	F_1^{q}	Invalid rate
Qwen-3B	RAGTAG	3	0.697	0.711	0.803	0.577	0.2%
	BRAGTAG	6	0.714	0.693	0.803	0.645	1.8%
Qwen-7B	RAGTAG	6	0.718	0.731	0.816	0.605	3.5%
	BRAGTAG	12	0.738	0.742	0.805	0.666	3.8%
Qwen-14B	RAGTAG	12	0.732	0.735	0.838	0.622	4.5%
	BRAGTAG	15	0.756	0.745	0.840	0.683	4.7%
Qwen-32B	RAGTAG	12	0.767	0.759	0.842	0.699	4.6%
	BRAGTAG	12	0.781	0.775	0.840	0.729	4.0%



■ **Figure 6** (a) Macro F_1 for RAGTAG at its best k in PS vs Fine-Tune in PS and PA across all four model sizes. (b) Per-class F_1 gain from Fine-Tune PS to Fine-Tune PA ($\Delta F_1 = \text{PA} - \text{PS}$) for each class and model size.

403 F_1 improves by 0.030–0.068 across all four model sizes. Bug F_1 stays within 0.018 of
 404 RAGTAG, with a slight loss for Qwen-3B and slight gains for the larger models. Therefore,
 405 removing bug neighbors in some cases did not decrease overall bug performance. Feature F_1
 406 is essentially unchanged. The question-to-bug misclassification rate drops: at each model’s
 407 best k , RAGTAG misclassifies 26–36% of true-question issues as bugs while BRAGTAG
 408 reduces this to 19–27%. Averaged across $k \in [1, 15]$, the rate drops from 31–40% (RAGTAG)
 409 to 24–36% (BRAGTAG). Question-to-feature misclassification is unaffected. This confirms
 410 that BRAGTAG’s improvement comes from the targeted reduction of question-to-bug
 411 misclassification.

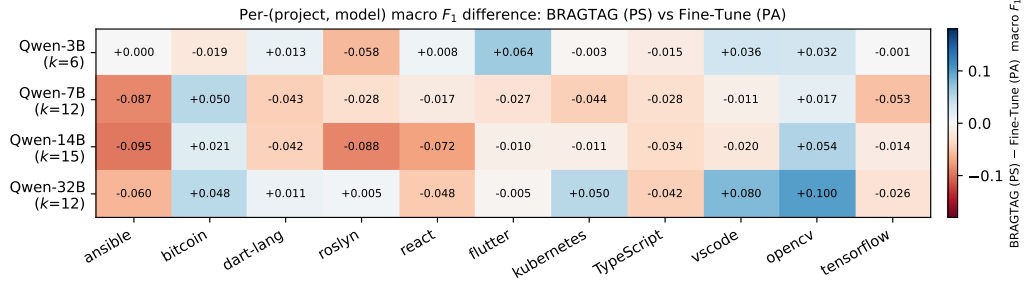
412 Outputs from both approaches that failed to parse to a valid label are marked as *invalid*
 413 and counted as incorrect. Averaged across $k \in [1, 15]$, BRAGTAG has a lower invalid rate
 414 than RAGTAG (2.3% vs 3.0% across the four models).

Answer to RQ3: The balanced retrieval method (BRAGTAG), outperforms RAGTAG at every $k \geq 6$. The gain comes from a targeted reduction of question-to-bug misclassifications, raising question F_1 by 0.030–0.068 without sacrificing bug or feature performance.

416 5.4 RQ4: Few-Shot Prompting vs. LoRA Fine-Tuning for IRC

■ **Table 2** RAGTAG, BRAGTAG, and Fine-Tune at each method’s best configuration (RAGTAG/BRAGTAG at their best k on PS, Fine-Tune on PA). Macro F_1 and per-class F_1 are computed on raw predictions. The +VOTAG column reports macro F_1 with VOTAG as fallback for the invalid LLM outputs. Train and Inference are run-times on a single GPU with Total as their sum. RAM is the peak GPU memory observed

Model	Method	k^*	Macro F_1	+VOTAG	F_1^{bug}	F_1^{feat}	F_1^{q}	RAM (GB)	Train (h)	Infer (h)	Total (h)
Qwen-3B	RAGTAG	3	0.697	0.696	0.711	0.803	0.577	2.9	–	0.18	0.18
	BRAGTAG	6	0.714	0.720	0.693	0.803	0.645	2.9	–	0.22	0.22
	Fine-Tune	–	0.708	0.711	0.736	0.790	0.599	5.5	0.31	0.11	0.42
Qwen-7B	RAGTAG	6	0.718	0.729	0.731	0.816	0.605	6.8	–	0.50	0.50
	BRAGTAG	12	0.738	0.751	0.742	0.805	0.666	6.8	–	0.60	0.60
	Fine-Tune	–	0.762	0.762	0.762	0.846	0.678	9.9	0.48	0.10	0.58
Qwen-14B	RAGTAG	12	0.732	0.746	0.735	0.838	0.622	12.6	–	1.37	1.37
	BRAGTAG	15	0.756	0.771	0.745	0.840	0.683	12.6	–	1.35	1.35
	Fine-Tune	–	0.785	0.785	0.792	0.828	0.736	16.7	1.49	0.22	1.71
Qwen-32B	RAGTAG	12	0.767	0.779	0.759	0.842	0.699	22.3	–	4.62	4.62
	BRAGTAG	12	0.781	0.792	0.775	0.840	0.729	22.3	–	4.15	4.15
	Fine-Tune	–	0.771	0.771	0.791	0.853	0.669	31.5	2.28	0.44	2.71



■ **Figure 7** Per-(project, model) macro F_1 difference between BRAGTAG (PS) and fine-tune (PA), across the 11 projects and four Qwen sizes. Each BRAGTAG row uses that model’s best k .

Classification Performance. We fine-tune all models in both PA and PS settings. Figure 6(a) compares macro F_1 for fine-tuning in both settings against RAGTAG’s PS at each model’s best k . Fine-tuning PA outperforms fine-tuning PS at every model size, by +0.033 (Qwen-3B), +0.084 (Qwen-7B), +0.082 (Qwen-14B), and +0.031 (Qwen-32B) macro F_1 . This trend matches prior findings on LLM fine-tuning for IRC [23].

RAGTAG PS outperforms Fine-Tune PS at every model size, by +0.021 (Qwen-3B) to +0.040 (Qwen-7B) macro F_1 . This shows that using only 300 labeled issues per project, retrieval-augmented few-shot is more effective than LoRA fine-tuning for IRC.

Figure 6(b) shows where the additional PA training data helps most in fine-tuning. Averaged across all four models, bug F_1 gains +0.070 going from PS to PA, question gains +0.053, and feature gains +0.050. Qwen-32B is the only exception: its question F_1 drops by 0.010 while bug gains +0.070.

Table 2 reports macro and per-class F_1 at each method’s best configuration. RAGTAG has competitive macro F_1 with fine-tuning: aggregated across the four model sizes (13,200 paired predictions), fine-tune leads RAGTAG by 0.029 macro F_1 (paired bootstrap 95% CI $[-0.037, -0.022]$). RAGTAG is closer to fine-tune at the smallest and largest models and farther in the middle, with per-model (RAGTAG – fine-tune) macro F_1 deltas (paired bootstrap 95% CIs in brackets) of -0.011 $[-0.028, +0.007]$ (Qwen-3B), -0.044 $[-0.059, -0.029]$ (Qwen-7B), -0.053 $[-0.068, -0.039]$ (Qwen-14B), and -0.004 $[-0.020, +0.009]$ (Qwen-32B). RAGTAG and fine-tune are statistically equal at Qwen-3B and Qwen-32B.

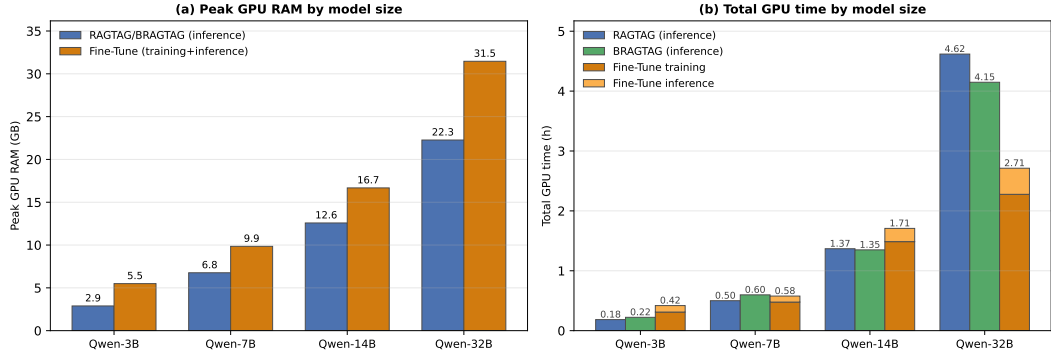


Figure 8 (a) Peak GPU RAM by model size: RAGTAG/BRAGTAG inference vs Fine-Tune training + inference. (b) Total GPU time by model size: RAGTAG and BRAGTAG vs Fine-Tune

BRAGTAG’s balanced retrieval narrows this gap: the aggregate (BRAGTAG – fine-tune) macro F_1 difference is -0.010 . However, fine-tuning does not consistently lead across projects. Figure 7 reports the per-(project, model) macro F_1 difference between BRAGTAG and fine-tune. BRAGTAG outperforms fine-tune in 16 of the 44 (project, model) cells, winning all four model sizes on `opencv` and three of four on `bitcoin`. Similar to RAGTAG, BRAGTAG performs better relative to fine-tune at the smallest and largest models: BRAGTAG is statistically tied with fine-tune at Qwen-3B and Qwen-32B while fine-tune leads on the mid-sized ones, with per-model (BRAGTAG – fine-tune) deltas of $+0.006$ $[-0.012, +0.024]$ (Qwen-3B), -0.024 $[-0.039, -0.009]$ (Qwen-7B), -0.029 $[-0.044, -0.013]$ (Qwen-14B), and $+0.010$ $[-0.005, +0.024]$ (Qwen-32B).

BRAGTAG has the best question F_1 on Qwen-3B and Qwen-32B, while fine-tune leads on Qwen-7B and Qwen-14B. Fine-tune leads on bug across all four model sizes; on feature, BRAGTAG wins Qwen-3B and Qwen-14B while fine-tune wins Qwen-7B and Qwen-32B. These results show that BRAGTAG is able to compete with fine-tune due to its targeted question-to-bug correction, boosting question F_1 without sacrificing bug or feature. Averaged across the four models, BRAGTAG is also the most class-balanced approach, with the lowest per-class F_1 standard deviation (0.058 vs 0.067 for fine-tune and 0.082 for RAGTAG) and the highest worst-class F_1 floor (0.681 vs 0.671 and 0.626).

Fine-tune produces fewer invalid outputs than RAGTAG and BRAGTAG in both settings, because the model learns the output format during training. Averaged across the four models, fine-tune’s invalid rate is 0.28% in PA and 0.38% in PS, below RAGTAG’s 3.0% and BRAGTAG’s 2.3% (averaged across $k \in [1, 15]$). In an ideal triage scenario, every issue report receives a label, so a known limitation of LLMs is the production of outputs that do not comply with the instructions [43, 2]. To compensate for the invalid predictions we propose using VOTAG as a fallback mechanism. VOTAG never produces an invalid label and uses the neighbors already retrieved for RAGTAG/BRAGTAG with no LLM inference cost. We apply the fallback to all three methods at their best data scopes for fair comparison: VOTAG-PS for RAGTAG/BRAGTAG and VOTAG-PA for fine-tune. Under this protocol, BRAGTAG’s reaches statistical equivalence with fine-tune, passing TOST at $\delta=0.01$ (bootstrap 95% CI $[-0.007, +0.008]$ ⁶).

⁶ 1,000 paired bootstrap resamples per model for both RAGTAG vs fine-tune and BRAGTAG vs fine-tune comparisons.

Computation and Data Cost. The methods have different hardware requirements. Figure 8(a) shows peak GPU RAM by model. RAGTAG and BRAGTAG require identical GPU memory that grows with model size, while fine-tune requires more at every size due to training overheads such as, gradients, optimizer state, and activations retained for the backward pass. Averaged across the model sizes, RAGTAG and BRAGTAG take 33% less GPU memory than fine-tune (47%, 31%, 25%, and 29% less at Qwen-3B, 7B, 14B, and 32B, respectively).

Figure 8(b) shows total GPU time at each model size. Fine-tune takes longer than RAGTAG at Qwen-3B, 7B, and 14B. But at Qwen-32B fine-tune (2.71 h) takes less time than both RAGTAG (4.62 h) and BRAGTAG (4.15 h). Inference time on each model naturally grows with prompt length. Fine-tune processes only the query issue, while RAGTAG and BRAGTAG process significantly more context due to the additional few-shot examples per query. As a result, fine-tune inference takes much less time than the other two methods, and fine-tune’s total run-time is dominated by training. At Qwen-7B and Qwen-32B, BRAGTAG’s inference cost alone exceeds fine-tune’s combined training and inference time. BRAGTAG’s balanced retrieval shortens the average context enough at Qwen-14B and Qwen-32B for BRAGTAG to run less than RAGTAG. However, BRAGTAG runs longer at Qwen-3B and Qwen-7B because the intervention could not compensate for the higher best k value.

Notably, RAGTAG and BRAGTAG in PS retrieve only from 300 labeled issues per project, while fine-tune PA trains on the full 3,300 issues across all 11 projects to reach its best configuration. Therefore, at $11\times$ less labeled data per project, RAGTAG remains competitive with fine-tune, and BRAGTAG closes the gap to TOST equivalence at $\delta=0.02$, tightening to statistical equivalence at $\delta=0.01$ under the VOTAG fallback for invalid LLM outputs.

Answer to RQ4: RAGTAG outperforms LoRA fine-tuning by 0.021–0.040 macro F_1 at every model size with project-specific data scope. Fine-tuning outperforms RAGTAG only when trained on $11\times$ more labeled data via cross-project pooling (PA), and leads by 0.029 aggregate macro F_1 . BRAGTAG narrows the aggregate gap to -0.010 , reaching statistical equivalence with fine-tuning once a VOTAG fallback handles invalid LLM outputs. Finally, Retrieval-augmented few-shot requires 33% less peak GPU memory than fine-tuning.

6 Discussion

6.1 Failure Analysis

In this section we conduct a manual qualitative analysis to better understand the misclassification patterns across the explored IRC methods. We inspect the misclassifications produced by Qwen-32B (the strongest model) under each method’s best configuration. From all the misclassifications across RAGTAG, BRAGTAG, and fine-tune, we drew a random sample of 50 misclassifications per method (150 total). Each Method’s sample includes 30 true-question, 10 true-bug, and 10 true-feature errors. The 30/10/10 split mimics the empirical class distribution of misclassifications pooled across the three methods. Each case was analyzed by two authors independently, with any discrepancies resolved through discussions. The observed failure categories and their frequencies are as follows:

Wrong template format (77.33%). Issue reports whose content and ground-truth label contradicts their chosen template and structure. All of these errors are from questions

and feature requests that the user filed under a bug-report template (*Steps to reproduce, Expected, Actual, System Information*). Consequently, the structure of the bug template misleads the LLM classifier towards predicting bug, regardless of the issue’s content. For example, [microsoft/vscode#193985](https://github.com/microsoft/vscode/issues/193985)⁷ is an actual support question that opens with a literal `Type:Bug` because it uses VS Code’s bug template.

Hybrid intent (13.33%). Issues that could have multiple defensible labels due to their combined intents. These errors mostly take place when a fix or a feature request is layered in a question or bug. For example, in [kubernetes/kubernetes#120364](https://github.com/kubernetes/kubernetes/issues/120364)⁸ the author both reports a concrete bug and in the same body explicitly requests a new feature: “*Please, give us multiarchitecture image! and fix monoarchitecture images too!*”

Label noise (9.33%). Issues whose ground-truth label does not match its content. For example, [tensorflow/tensorflow#61710](https://github.com/tensorflow/tensorflow/issues/61710)⁹ is titled “*Will there a MLP model in the future version?*” which requests a new API feature for building MLPs. Yet the dataset labels it as a bug.

We also sampled 30 issues with *invalid* predictions per method (60 total) and observed three failure modes. In roughly 73% of cases, the model either continued to generate issue-body content or its own reasoning. Another 23% were hallucinations such as repeated digits and punctuations. The remaining 3% are parsable labels that fall outside the three valid classes (e.g., `documentation`).

6.2 Key Insights and Practical Implications

The practicality of context retrieval for IRC. Our results indicate that RAG-based approaches offer a more practical solution for IRC than fine-tuning. With $11\times$ less labeled data per project and 25–47% less peak GPU memory, RAGTAG performs competitively against fine-tuning (trails by 0.029 in aggregate macro F1), and BRAGTAG closes the gap to TOST equivalence at $\delta=0.02$, tightening to statistical equivalence at $\delta=0.01$ under the VOTAG fallback mechanism for invalid LLM outputs (Section 5.4). RAG-based methods offer deployment advantages over fine-tuning as well: adding a new labeled issue report becomes an incremental index update rather than a retraining cycle, and the model can be swapped in production with no project-specific retraining. For extremely resource-constrained scenarios, even VOTAG PS is a defensible baseline: it reaches macro $F_1 = 0.595$ ($k=15$), within 0.018 of Qwen-3B’s zero-shot 0.613 (Section 5.1, Section 5.2), at a fraction of the GPU cost (≈ 240 MB vs 2.9 GB).

Combining classification and validation. Our failure analysis (Section 6.1) shows that users frequently file support questions and feature requests under the bug-report template, which can bias the classifier towards predicting bug regardless of the content. This emphasizes the importance of refining the new LLM-assisted triage pipelines now in place on platforms such as GitHub [20] and GitLab [21]. In these pipelines, the reporter describes the report to an LLM, which then suggest the correct template and labels. Report validation is also important since mislabeling and duplication is a common problem among contributors[3, 40]. Studies have already explored validation for bug reports [32, 49, 1, 16]. However, combining classification and validation in a cohesive workflow is underexplored.

⁷ <https://github.com/microsoft/vscode/issues/193985>

⁸ <https://github.com/kubernetes/kubernetes/issues/120364>

⁹ <https://github.com/tensorflow/tensorflow/issues/61710>

6.3 Future Work

Generalization to more diverse environments. Our analysis covers a large benchmark across 11 GitHub projects with a balanced 600 issues per project. We encourage future work to evaluate IRC methods under more diverse settings such as imbalanced label distributions with one or more minority classes, multi-label settings where more than one label can be predicted for a single issue, and other issue tracking platforms (e.g., Jira, Bugzilla). These future directions will assess the generalizability of our findings and uncover new patterns.

Other retrieval techniques. Our study empirically shows that retrieval heuristics can soften model biases and even match fine-tuning without parameter updates. This opens the door for future research on other retrieval-side interventions for different IRC settings. A concrete future direction is a two-stage inference scheme where a first LLM call filters or re-ranks the retrieved neighbors based on their relevance, and a second call performs the classification. Settings such as embedding model and similarity metric are also worth exploring in future work, because of their impact on retrieval quality.

Prompt tuning Our proposed few-shot prompt structure is one of many reasonable options. Since other formats can yield better results, we encourage future work to explore other structures and advanced strategies such as Chain-of-Thought (CoT) prompting [33].

Other LLM families. Our study uses the Qwen2.5-Instruct family at four sizes to isolate model scale from architecture (Section 4). Now that the competitiveness of retrieval-augmented few-shot prompting is empirically established on this family, we encourage future work to conduct a dedicated study identifying which open-source LLM families are most effective for retrieval-augmented few-shot IRC and characterizing how architectural choices affect retrieval-augmented classification.

7 Threats to Validity

Internal Validity. Empirical selection of BRAGTAG’s margin can potentially be a threat to internal validity, since the value could be specifically tuned for the models and test data in this study. To mitigate this, margin calibration used only the training split and did not involve the test split (Section 5.2). Furthermore, the same fixed margin is used across all models and projects with no per-project or per-model tuning. Finally, the selected margin in this study is not framed as a universally optimal constant. Any project can re-derive the margin with negligible cost by only using the retrieval index of available historical issue reports. The choice of hyperparameters and prompt format can also pose an internal threat to validity. Different hyperparameters and prompts can change model performance. We address this when reproducing the fine-tuning baseline in this study by adopting the same hyperparameters and prompt formats used in prior work [23, 5].

External Validity. One threat to external validity concerns the generalizability of the results. Issue report text can vary substantially in writing style, vocabulary, and structure across software projects [2], which may threaten the generalizability of our findings. To mitigate this threat, we used a large benchmark of 6,600 issues from 11 different projects, each with its own reporting conventions and writing styles. Another threat to external validity is the generalizability across settings. The primary aim of this work is to empirically assess the effectiveness of retrieval-augmented few-shot prompting versus LoRA fine-tuning for IRC and to explore how results vary with different model scales. This goal required exploring a large variable space across model sizes, k values, data scopes, and different

retrieval-based methods. However, Many configurations, including LLM architecture, prompt structure, embedding model, and retrieval technique, can each lead to different results, which can be a potential threat to the generalizability of our findings beyond the settings used in this study. Recognizing this threat, we encourage future work to build on our findings and conduct dedicated studies under different settings. To facilitate such studies, we designed our experiment pipeline in a modular way that allows all configurations to be easily substituted.

Construct Validity. The inherent randomness of LLM outputs can pose a potential threat to construct validity. The reported performance metrics can vary between different runs due to LLM randomness. To mitigate this variance, we ran the experiments three times and report the average.

Reliability. [Provide the link to replication package here.]

8 Conclusion

[AH: This is actually the abstract. Needs to be rewritten.] We evaluate all three against a LoRA fine-tuning baseline on an 11-project benchmark, in both *project-specific* (PS) and *project-agnostic* (PA) data scope settings. Experiments use Qwen2.5-Instruct at four sizes. In this study, we investigated retrieval-augmented few-shot prompting as a cost-efficient alternative to LoRA fine-tuning for issue report classification (IRC). Given a query issue, we retrieved its top- k nearest neighbors from historical issue reports to be used in three proposed methods. First, RAGTAG, which presents the neighbors as labeled examples in a few-shot prompt. Second, BRAGTAG, a RAGTAG variant that drops bug-labeled neighbors when the retrieved label distribution indicates the query is likely a question. Third, VOTAG, which predicts the label by a weighted vote on the neighbors' labels, establishing a non-parametric baseline.

We evaluated all three methods against the LoRA fine-tuning baseline using the Qwen2.5-Instruct model at four sizes. Evaluations were conducted under both project-specific (PS) and project-agnostic (PA) data scope settings. The results show that retrieval-augmented few-shot prompting is a practical alternative to fine-tuning for IRC. RAGTAG PS outperforms fine-tuning PS by 0.021–0.040 macro F_1 across all four model sizes, and trails fine-tuning PA by 0.029 macro F_1 in aggregate despite using $11\times$ less labeled data per project. BRAGTAG PS closes this gap to TOST equivalence at $\delta=0.02$, tightening to statistical equivalence at $\delta=0.01$ under the VOTAG fallback protocol for invalid LLM outputs. Averaged across all models, retrieval-augmented few-shot also required 33% less peak GPU memory than fine-tuning.

These findings position retrieval-augmented few-shot prompting as the more practical IRC method, particularly when hardware is constrained, labeled data is limited, or deployment requires frequent model or data updates. We encourage future work to evaluate these methods in diverse data distribution scenarios and to explore additional retrieval techniques and configurations.

Data Availability

[Our replication package [?]] contains the code, data, and all necessary materials to reproduce the results presented in this paper.

References

- 1 Toufique Ahmed, Jatin Ganhotra, Rangeet Pan, Avraham Shinnar, Saurabh Sinha, and Martin Hirzel. Otter: Generating tests from issues to validate swe patches. *arXiv preprint arXiv:2502.05368*, 2025.
- 2 Arshia Akhavan, Alireza Hoseinpour, Abbas Heydarnoori, Hamid Bagheri, and Mehdi Keshani. Linkanchor: An autonomous llm-based agent for issue-to-commit link recovery, 2026. URL: <https://arxiv.org/abs/2508.12232>, arXiv:2508.12232.
- 3 Giuliano Antoniol, Kamel Ayari, Massimiliano Di Penta, Foutse Khomh, and Yann-Gaël Guéhéneuc. Is it a bug or an enhancement? a text-based approach to classify change requests. In *Proceedings of the 2008 conference of the center for advanced studies on collaborative research: meeting of minds*, pages 304–318, 2008.
- 4 Gabriel Aracena, Kyle Luster, Fabio Santos, Igor Steinmacher, and Marco A. Gerosa. Applying large language models api to issue classification problem, 2024. URL: <https://arxiv.org/abs/2401.04637>, arXiv:2401.04637.
- 5 Gabriel Aracena, Kyle Luster, Fabio Santos, Igor Steinmacher, and Marco A Gerosa. Applying large language models to issue classification: Revisiting with extended data and new models. *Science of Computer Programming*, page 103333, 2025.
- 6 Maram Assi, Safwat Hassan, and Ying Zou. Llm-cure: Llm-based competitor user review analysis for feature enhancement. *ACM Transactions on Software Engineering and Methodology*, 35(4):1–25, 2026.
- 7 Shikhar Bharadwaj and Tushar Kadam. Github issue classification using bert-style models. In *Proceedings of the 1st International Workshop on Natural Language-based Software Engineering*, pages 40–43, 2022.
- 8 Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- 9 Jordi Cabot, Javier Luis Cánovas Izquierdo, Valerio Cosentino, and Belén Rolandi. Exploring the use of labels to categorize issues in open-source software projects. In *2015 IEEE 22nd International Conference on Software Analysis, Evolution, and Reengineering (SANER)*, pages 550–554. IEEE, 2015.
- 10 Gemma Catolino, Fabio Palomba, Andy Zaidman, and Filomena Ferrucci. Not all bugs are the same: Understanding, characterizing, and classifying bug types. *Journal of Systems and Software*, 152:165–181, 2019.
- 11 Giuseppe Colavito, Filippo Lanubile, and Nicole Novielli. Issue report classification using pre-trained language models. In *Proceedings of the 1st International Workshop on Natural Language-based Software Engineering*, pages 29–32, 2022.
- 12 Giuseppe Colavito, Filippo Lanubile, and Nicole Novielli. Benchmarking large language models for automated labeling: The case of issue report classification. *Information and Software Technology*, page 107758, 2025.
- 13 Giuseppe Colavito, Filippo Lanubile, Nicole Novielli, and Luigi Quaranta. Large language models for issue report classification. In *Ital-IA*, pages 18–23, 2024.
- 14 Giuseppe Colavito, Filippo Lanubile, Nicole Novielli, and Luigi Quaranta. Leveraging gpt-like llms to automate issue labeling. In *Proceedings of the 21st International Conference on Mining Software Repositories*, pages 469–480, 2024.
- 15 Thomas Cover and Peter Hart. Nearest neighbor pattern classification. *IEEE transactions on information theory*, 13(1):21–27, 1967.
- 16 Emre Dinç and Eray Tüzün. Judge the votes: A system to classify bug reports and give suggestions. In *2025 2nd IEEE/ACM International Conference on AI-powered Software (AIware)*, pages 01–10. IEEE, 2025.

- 681 17 Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-
682 Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. The faiss library. *IEEE*
683 *Transactions on Big Data*, 2025.
- 684 18 Sahibsingh A Dudani. The distance-weighted k-nearest-neighbor rule. *IEEE Transactions on*
685 *Systems, Man, and Cybernetics*, (4):325–327, 1976.
- 686 19 Qiang Fan, Yue Yu, Gang Yin, Tao Wang, and Huaimin Wang. Where is the road for issue
687 reports classification based on text mining? In *2017 ACM/IEEE International Symposium on*
688 *Empirical Software Engineering and Measurement (ESEM)*, pages 121–130. IEEE, 2017.
- 689 20 GitHub. GitHub Copilot. <https://github.com/features/copilot>, 2025.
- 690 21 GitLab. GitLab Duo. <https://about.gitlab.com/gitlab-duo/>, 2025.
- 691 22 Luiz Gomes, Ricardo da Silva Torres, and Mario Lúcio Côrtes. Bert-and tf-idf-based feature
692 extraction for long-lived bug prediction in floss: A comparative study. *Information and*
693 *Software Technology*, 160:107217, 2023.
- 694 23 Jueun Heo and Seonah Lee. A study on applying large language models to issue classification.
695 In *2025 IEEE/ACM 33rd International Conference on Program Comprehension (ICPC)*, pages
696 1–11. IEEE Computer Society, 2025.
- 697 24 Kim Herzig, Sascha Just, and Andreas Zeller. It’s not a bug, it’s a feature: how misclassification
698 impacts bug prediction. In *2013 35th international conference on software engineering (ICSE)*,
699 pages 392–401. IEEE, 2013.
- 700 25 Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang
701 Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3,
702 2022.
- 703 26 Huihui Huang, Ratnadira Widyasari, Ting Zhang, Ivana Clairine Irsan, Jieke Shi, Han Wei
704 Ang, Frank Liauw, Eng Lieh Ouh, Lwin Khin Shar, Hong Jin Kang, et al. Back to the basics:
705 Rethinking issue-commit linking with llm-assisted retrieval. *arXiv preprint arXiv:2507.09199*,
706 2025.
- 707 27 Robert L. Logan IV, Ivana Balažević, Eric Wallace, Fabio Petroni, Sameer Singh, and Sebastian
708 Riedel. Cutting down on prompts and parameters: Simple few-shot learning with language
709 models, 2021. URL: <https://arxiv.org/abs/2106.13353>, arXiv:2106.13353.
- 710 28 Maliheh Izadi. Catiss: An intelligent tool for categorizing issues reports using transformers. In
711 *Proceedings of the 1st international workshop on natural language-based software engineering*,
712 pages 44–47, 2022.
- 713 29 Maliheh Izadi, Kiana Akbari, and Abbas Heydarnoori. Predicting the objective and priority
714 of issue reports in software repositories. *Empirical Software Engineering*, 27(2):50, 2022.
- 715 30 Gaetul Jeong, Sunghun Kim, and Thomas Zimmermann. Improving bug triage with bug
716 tossing graphs. In *Proceedings of the 7th joint meeting of the European software engineering*
717 *conference and the ACM SIGSOFT symposium on The foundations of software engineering*,
718 pages 111–120, 2009.
- 719 31 Rafael Kallis, Andrea Di Sorbo, Gerardo Canfora, and Sebastiano Panichella. Ticket tagger:
720 Machine learning driven issue classification. In *2019 IEEE International Conference on*
721 *Software Maintenance and Evolution (ICSME)*, pages 406–409. IEEE, 2019.
- 722 32 Lara Khatib, Noble Saji Mathews, and Meiyappan Nagappan. Assertflip: Reproducing bugs
723 via inversion of llm-generated passing tests. *arXiv preprint arXiv:2507.17542*, 2025.
- 724 33 Anil Koyuncu. Exploring fine-grained bug report categorization with large language models
725 and prompt engineering: An empirical study. *ACM Transactions on Software Engineering*
726 *and Methodology*, 2025.
- 727 34 Van-Hoang Le and Hongyu Zhang. Log parsing with prompt-based few-shot learning. In *2023*
728 *IEEE/ACM 45th International Conference on Software Engineering (ICSE)*, pages 2438–2449.
729 IEEE, 2023.
- 730 35 Zhifang Liao, Dayu He, Zhijie Chen, Xiaoping Fan, Yan Zhang, and Shengzong Liu. Exploring
731 the characteristics of issue-related behaviors in github using visualization techniques. *IEEE*
732 *Access*, 6:24003–24015, 2018.

- 36 Jiachang Liu, Dinghan Shen, Yizhe Zhang, William B Dolan, Lawrence Carin, and Weizhu Chen. What makes good in-context examples for gpt-3? In *Proceedings of Deep Learning Inside Out (DeeLIO 2022): The 3rd workshop on knowledge extraction and integration for deep learning architectures*, pages 100–114, 2022.
- 37 Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics*, 12:157–173, 2024.
- 38 Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- 39 Huan Ma, Changqing Zhang, Yatao Bian, Lema Liu, Zhirui Zhang, Peilin Zhao, Shu Zhang, Huazhu Fu, Qinghua Hu, and Bingzhe Wu. Fairness-guided few-shot prompting for large language models, 2023. URL: <https://arxiv.org/abs/2303.13217>, arXiv:2303.13217.
- 40 Nitish Pandey, Debarshi Kumar Sanyal, Abir Hudait, and Amitava Sen. Automated classification of software issue reports using machine learning techniques: an empirical study. *Innovations in Systems and Software Engineering*, 13(4):279–297, 2017.
- 41 Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*, 2019.
- 42 Fabio Santos, Joseph Vargovich, Bianca Trinkenreich, Italo Santos, Jacob Penney, Ricardo Britto, João Felipe Pimentel, Igor Wiese, Igor Steinmacher, Anita Sarma, et al. Tag that issue: applying api-domain labels in issue tracking systems. *Empirical Software Engineering*, 28(5):116, 2023.
- 43 Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models’ sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. *arXiv preprint arXiv:2310.11324*, 2023.
- 44 Igor Steinmacher, Christoph Treude, and Marco Aurelio Gerosa. Let me in: Guidelines for the successful onboarding of newcomers to open source projects. *IEEE Software*, 36(4):41–49, 2018.
- 45 Alexander Trautsch and Steffen Herbold. Predicting issue types with sebert. In *Proceedings of the 1st international workshop on natural language-based software engineering*, pages 37–39, 2022.
- 46 Joseph Vargovich, Fabio Santos, Jacob Penney, Marco A Gerosa, and Igor Steinmacher. Givemelabeledissues: An open source issue recommendation system. In *2023 IEEE/ACM 20th International Conference on Mining Software Repositories (MSR)*, pages 402–406. IEEE, 2023.
- 47 Julian Von der Mosel, Alexander Trautsch, and Steffen Herbold. On the validity of pre-trained transformers for natural language processing in the software engineering domain. *IEEE Transactions on Software Engineering*, 49(4):1487–1507, 2022.
- 48 Jun Wang, Xiaofang Zhang, and Lin Chen. How well do pre-trained contextual language representations recommend labels for github issues? *Knowledge-Based Systems*, 232:107476, 2021.
- 49 Xinchun Wang, Pengfei Gao, Xiangxin Meng, Chao Peng, Ruida Hu, Yun Lin, and Cuiyun Gao. Aegis: An agent-based framework for general bug reproduction from issue descriptions. *arXiv preprint arXiv:2411.18015*, 2024.
- 50 Hongrun Wu, Yutao Ma, Zhenglong Xiang, Chen Yang, and Keqing He. A spatial-temporal graph neural network framework for automated software bug triaging. *Knowledge-Based Systems*, 241:108308, 2022.
- 51 An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- 52 Guoxin Yu, Lema Liu, Haiyun Jiang, Shuming Shi, and Xiang Ao. Retrieval-augmented few-shot text classification. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6721–6735, 2023.